

MLP OPPE 2 - Classification Practice Paper

(Student Performance Domain)

PreProcessing Dataset: preprocessing_student_classification.csv

Q1 [NAT] Find ratio of missing values in attendance to non-missing values in department. Round to 2 decimals.

Q2 [MCQ] Which pair has the least absolute correlation coefficient? Options: attendance-backlogs, study_hours-final_score, attendance-final_score, assignment_score-midterm_score

Q3 [NAT] Find absolute difference between average final_score of Online and Offline students.

Q4 [NAT] Count rows satisfying all: study_mode is Online or Hybrid, attendance is not null, final_score > median final_score of same study_mode, study_hours < department-wise 65th percentile study_hours, backlogs < overall median backlogs.

Q5 [MCQ] After imputing numerical with mean and categorical with most frequent value, which has highest median? Options: attendance, assignment_score, study_hours, final_score, backlogs

Q6 [NAT] After MinMaxScaler on attendance, study_hours, backlogs and StandardScaler on assignment_score, final_score, compute $\text{Var}(\text{study_hours})/\text{Std}(\text{final_score})$, round to 3 decimals.

Q7 [NAT] One Hot Encode department (remove original column) and Ordinal Encode study_mode. What is the difference in number of columns between new and old dataframe?

Q8 [MCQ] After all preprocessing, closest mean of attendance is: 0.47, 0.53, 0.58, 0.62

Model Building Dataset: model_building_student_classification.csv

Target column = placement_selected. Use 80:20 split with random_state=0.

Q9 [NAT] Train LogisticRegression with C=0.5 and max_iter=3000. Find accuracy on test set.

Q10 [NAT] Find coefficient of attendance from LogisticRegression rounded to 2 decimals.

Q11 [NAT] Train RandomForestClassifier with n_estimators=200, max_depth=8, random_state=0. Compute recall score on test set.

Q12 [MCQ] Which feature gets least feature importance in RandomForest? Options: attendance, backlogs, study_hours, final_score, assignment_score

Q13 [NAT] Train DecisionTreeClassifier with max_depth=6, min_samples_split=10, random_state=0. Find total number of nodes.

Q14 [MCQ] Perform GridSearchCV on KNeighborsClassifier with n_neighbors=[3,5,7,9], weights=['uniform','distance'], cv=5, scoring='f1'. Which combination is best?

Q15 [NAT] Train SVC with kernel='rbf', C=1.5 and gamma='scale'. Compute precision score on test set rounded to 2 decimals.